The Distributional Effects of AI's Productivity Gains: A Data Fusion Approach

1. Research question

Which occupations and tasks gain most from AI, and how reliably can the *distribution* of those gains be measured? The economy-wide average is policy-relevant, but it is not enough: any forecast, targeting rule, or workforce-transition plan also needs to know which jobs are affected, by how much, and with what confidence, and the average is silent on who gains and who does not. The object of this project is therefore the map of gains across the economy: a calibrated estimate for every cell, displayed cell by cell, in the spirit of the Opportunity Atlas (Chetty et al., 2025).

Formally, let $j = 1, \ldots, J$ index O*NET occupation-task cells, with J on the order of one thousand, and let θ_j be the productivity gain in cell j. The primitive is high-dimensional: the vector $\theta = (\theta_1, \ldots, \theta_J)$. Aggregated decisions often turn on a few low-dimensional summaries of θ, each requiring a credible uncertainty statement: the economy-wide aggregate $\Theta = \sum_j \omega_j \theta_j$, the cross-cell dispersion of gains, and top-k rankings or targeting sets. These summaries are not separately estimable, however: each is a functional of the entire vector, so recovering them credibly means getting the full map right.

Moreover, since AI also changes *which* tasks people do, the map comes in two versions: the gain on the tasks workers did before AI, and the gain on the tasks users actually bring to AI (Cunningham and Whitfill, 2026). The deliverable is the calibrated atlas and its decision-relevant summaries, in both basket versions, together with the result that the true value gain is bounded between the two; Figure 1 previews the deliverable on simulated data.

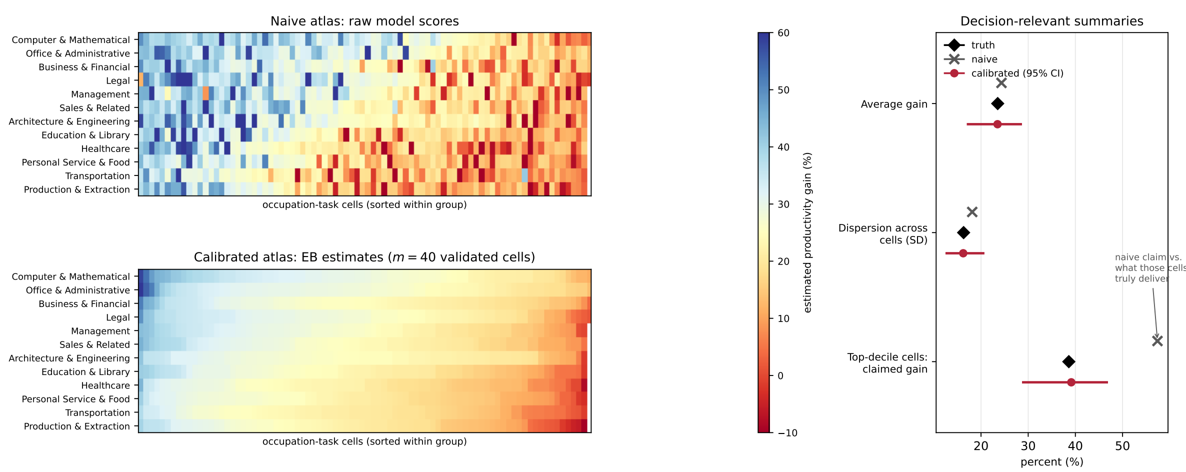

Figure 1: The deliverable, previewed on simulated data. Left: the atlas. The raw model scores (top) scatter noise across cells and compress true differences; the calibrated estimates (bottom, from $m = 40$ randomized validation cells) recover the real structure of who gains. Right: the decision-relevant summaries with uncertainty. The calibrated estimates recover the average, the cross-cell dispersion, and an honest top-decile gain with 95% intervals, while the naive spread mixes compressed signal with measurement noise, so its dispersion number is wrong in a direction that is not even interpretable (here it overshoots), and its claimed top-decile gain far exceeds what its selected cells truly deliver.

2. Motivation

Two literatures measure AI's productivity effect, and each is missing something the other has.

The first estimates time savings from AI usage data at scale: it classifies conversations into occupation-task cells, asks a model how long each task would take with and without AI, and aggregates the implied savings (Tamkin and McCrory, 2025). This is scalable, privacy-preserving inside the platform, and tracks real usage, but it has two weaknesses. One is measurement bias: the model's score S_j is a biased reading of the true gain θ_j, compressed toward the middle: genuine cross-cell differences are shrunk while measurement noise masquerades as spread, so the average survives but the dispersion that determines which occupations gain most is garbled, and the scores are then aggregated without correction. The other is an estimand problem: observed prompts are a selected sample of work, so uncorrected usage measures uplift on the tasks people already bring to AI, not uplift on the pre-AI task basket. The target is ambiguous until the basket is fixed.

The second uses experimental or quasi-experimental variation to measure genuine causal effects: customer-support agents (Brynjolfsson et al., 2025), software developers (Peng et al., 2023), management consultants (Dell'Acqua et al., 2026), and similar field studies. These are credibly identified, but each covers a single occupation or firm. They cannot produce an economy-wide map, and hand-picking a wider set of partner firms does not fix this: partnerships assembled through personal networks cannot span the occupation space in an unbiased way, so their gains do not generalize to the cells no one ran an experiment on.

Neither approach alone yields a credible map: the first has coverage but bias, the second has credibility but no coverage. The opening is a method that combines a broad but biased signal with a small, *deliberately randomized* experimental anchor, and uses the anchor to discipline the signal everywhere. Supplying that method is the project.

3. Empirical strategy

Inputs. (i) Usage data classifiable to occupation-task cells, with per-cell interaction counts and the model's own gain scores: broad coverage, but biased. (ii) A small ground-truth sample: a modest number of cells, chosen at random, in which the gain is measured directly by randomizing AI access at the worker-task level and recording completion time together with independently graded output quality.

The validation design. On the order of ten enterprise deployments, spanning occupation clusters, are drawn *at random* from the target population; within each, AI access or model tier is randomized at the worker or team level. Each deployment spans several occupation-task cells, so on the order of ten deployments supply the roughly forty validated cells that the budget below calls for. Random selection of the validation cells, before their gains are seen, is what makes the anchor represent the economy rather than the partners, so the experimental estimate $\widehat{\theta}_j^v$ is unbiased for θ_j and independent of it. Where some cells cannot be validated even in principle, conclusions for them rest on a stated extrapolation assumption and are reported as bounds.

The estimator. Treat the score as a biased, noisy measurement and recover the map by empirical Bayes. The validation pairs identify the score-to-truth channel, which is common to all cells; the scores then supply information for every cell, and posterior means combine the two, shrinking toward the calibrated score where a cell's own signal is noisy. The key saving is that the channel is a small, shared object, so the number of validated cells needed does not grow with J (see Appendix, A.3).

Simulation. Figure 2 shows the procedure on simulated data calibrated to the Tamkin–McCrory score quality (score-truth correlation ≈ 0.44).

To make the budget concrete, measure gains in percentage points; the accuracies that follow are root-mean-squared errors in those units. At current score quality, roughly forty randomly-chosen validated cells pin down both the economy-wide average gain and the *spread* of gains across

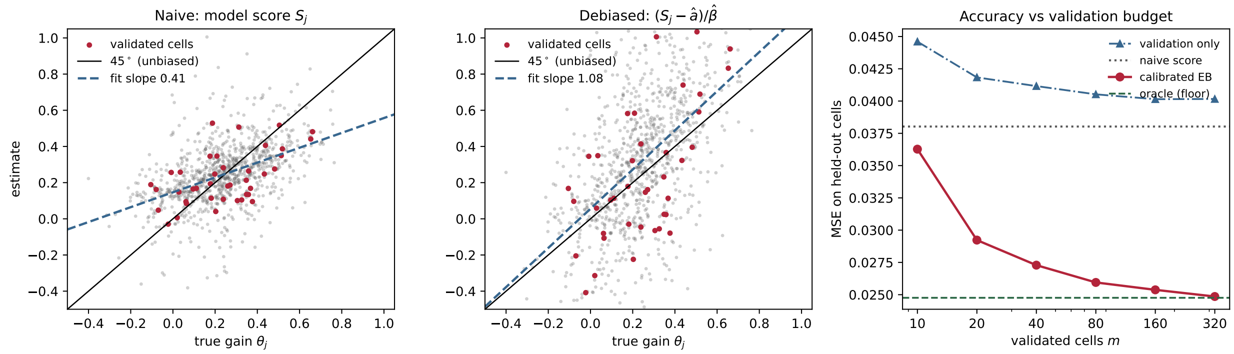

Figure 2: The raw score (left) compresses gains toward the middle: fitted slope 0.4, well below the 45° line, so it understates how much the most-affected occupations gain. Dividing by the validation-learned channel (middle) removes the compression (slope ≈ 1) at the cost of noise. Empirical-Bayes shrinkage then disciplines that noise (right): compound error on held-out cells falls toward the oracle, reaching its knee near $m \approx 40$ validated cells. Validation alone can only fall back on the population average for a cell it never measured, so it stays *above* the naive score at every budget.

occupations to within a few percentage points, and that spread is exactly what the naive score garbles: compression folds the forty-point gap between a cell where AI raises productivity by ten percent and one where it raises it by fifty into a sixteen-point difference in scores, while measurement noise fills the spread back in with error. A simulation study confirms this budget is robust even when the channel is strongly nonlinear. The exact budget-to-accuracy tradeoff is itself an output of the project.

4. Expected contribution

The *conceptual* contribution is to make the productivity estimand explicit. AI gains are not a scalar primitive; they depend on the task basket. The project therefore estimates two basket-specific maps and bounds value uplift between them, rather than reporting one number that hides a basket assumption.

The *methodological* contribution is an estimator for a high-dimensional map of latent cell-level effects, built from broad biased scores and a small randomized anchor, with the anchor's necessity, sufficiency, and cost made precise (Appendix). This extends prediction-powered inference and its many-mean shrinkage form (Angelopoulos et al., 2023; Li and Ignatiadis, 2025) from representative labels for a mean to a randomized anchor for a full latent vector; existing generated-data corrections target a single coefficient or bound a single scalar (Battaglia et al., 2025; Ludwig et al., 2025; Chen and Tamer, 2026).

The *substantive* contribution is the atlas itself, which reports not only the average gain but which occupations drive it, how uncertain the ranking is, and how much the calibrated picture revises the naive self-report baseline in its average, its dispersion, and its ranking of the most-affected cells.

The project falls under the broader category of measuring value: the estimand is task-level time savings and quality-adjusted output, aggregated into durable productivity gains, with the measurement calibrated rather than taken from model self-reports. It is executable only at the scale of OpenAI's usage data and with an OpenAI-supported validation design. It is privacy-preserving by construction: all conversation-level processing stays inside OpenAI's environment, and only cell-level aggregates above disclosure thresholds leave.

Appendix: formal statements

Statements only; proofs are in the accompanying research note, available on request. Fix one basket. True gains $\theta_j \sim N(\mu, \sigma_\theta^2)$ across cells $j = 1, \ldots, J$; the model score is $S_j = a + \beta\theta_j + \nu_j$ with $\nu_j \sim N(0, \sigma_S^2)$ and compression $0 < \beta < 1$; the validation estimate at selected cells is $\widehat{\theta}_j^v = \theta_j + \eta_j$, $\eta_j \sim N(0, s_j^2)$ with design-known s_j^2; selection into validation satisfies $R_j \perp \theta_j$ by randomization.

A.1 (Scores alone do not identify scale or level). The marginal law of $\{S_j\}$ depends on the parameters only through $a + \beta\mu$ and $\beta^2\sigma_\theta^2 + \sigma_S^2$; a one-parameter family of models with different dispersions σ_θ^2 and levels is observationally equivalent. Scores fix the ordering of cells but neither the scale nor the level of the map, which is exactly what aggregates and targeting conclusions require.

A.2 (The randomized anchor identifies everything). The validated pairs $\{(S_j, \widehat{\theta}_j^v) : R_j = 1\}$ identify $(\mu, \sigma_\theta^2, a, \beta, \sigma_S^2)$ by moment matching, and hence the posterior-mean rule, the aggregate, the dispersion, and the population ranking. Randomization is what makes the validated moments the population moments.

A.3 (Near-oracle accuracy from $m \ll J$). With the five parameters estimated from m validated cells, the feasible map's excess risk over the oracle's is $\Lambda_j/m + o(m^{-1})$ per cell, with Λ_j a computable sensitivity that does not depend on J; the aggregate error is $O_p(m^{-1/2})$, also free of J. This grounds the forty-cell benchmark of Section 3; the flexible-channel analogue is a research output of the project.

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Alex Tamkin and Peter McCrory. Estimating AI productivity gains from Claude conversations. Anthropic research report, 5 November 2025, 2025. URL https://www.anthropic.com/research/estimating-productivity-gains. Prompts Claude for time-with and time-without AI across 100,000 Claude.ai conversations; median ≈80% task time savings; extrapolated to ≈1.8% annual US labor-productivity growth. Validated against 1,000 tracked JIRA software tasks: Claude Sonnet 4.5 attains Spearman $\rho \approx 0.44$ (human developers 0.50), estimates compressed toward the mean (overestimates short tasks, underestimates long); 10-shot in-context calibration did not improve ρ. The validation set exists but the aggregated O*NET task $\times$ occupation estimates are used uncorrected.